

Neon AI Factory Architect

AI infrastructure architecture, engineering assurance and customer-ready design documentation

Neon AI Factory Architect (AIFactoryCAD) is a software platform for designing large-scale NVIDIA GPU infrastructure. It converts workload and deployment requirements into a governed, traceable AI-factory architecture covering compute, GPU fabrics, storage, networking, security, delivery planning and technical documentation.

Designed for **CTOs, Solution Architects, infrastructure engineering teams and technology executives**, the product combines deterministic engineering rules with curated vendor knowledge so that architecture decisions can be reproduced, reviewed and explained rather than generated as opaque AI recommendations.

Core capabilities

Capability	What the customer gets
Reference Architecture Compiler	Deterministic GPU infrastructure architectures from requirements, scale and deployment constraints, with immutable revision history and traceability.
NVIDIA GPU platforms	Current support for DGX GB300 NVL72 and HGX B300 architectures, including large-scale multi-rack deployments. Additional NVIDIA platforms are being progressively onboarded.
Accelerator fabrics	First-class modelling of NVLink/NVSwitch scale-up fabrics separately from Ethernet or InfiniBand scale-out networks, including GPU-domain sizing and topology semantics.
AI networking	Architecture support for NVIDIA Spectrum-X Ethernet and InfiniBand designs, with leaf/spine roles, port/link requirements and large-scale topology calculations.
Storage architecture	Governed storage sizing and architecture paths for VAST Data, WEKA and DDN , including capacity, throughput and architecture-specific evidence where available.

Optical connectivity	Deterministic derivation of optical-channel requirements, speeds, form factors, media, reach and endpoint compatibility.
Vendor Options Compiler	Presents selected products, compatible implementation options, architecture alternatives and evidence gaps without turning the tool into a procurement or ranking engine.
Security Architecture Compiler	Derives security zones, obligations, controls, evidence requirements and architecture-security relationships from the selected design.
Architecture Risk Compiler	Produces architecture-linked risk statements and severity assessments covering material design and delivery risks.
Project Delivery Planning	Generates structured implementation phases, workstreams, dependencies, gates, symbolic schedules and optional indicative project schedules.
Architecture documentation	Generates customer-reviewable JSON, Markdown, DOCX and PDF, SVG, Mermaid architecture documents with requirements traceability, risks, diagrams and design evidence.
Architecture visualisation	Multi-view architecture diagrams covering compute, network, storage, security and other domains, with offline rendering support.
Workbench & API	Browser-based architecture workflow plus authenticated APIs for controlled automation, revision management and downstream integration.

Engineering approach

Deterministic and evidence-backed

Architectures are produced from governed CKP/EKB knowledge packages, explicit compiler rules and vendor evidence, not free-form generative reasoning.

Revision-controlled

Every architecture, downstream analysis and document is bound to an immutable architecture revision and digest. Historical results remain auditable.

Transparent uncertainty

Missing sizing data or product evidence is surfaced explicitly as *unresolved*, *evidence required* or *architecture branch required* rather than silently guessed.

Architecture authority remains controlled

Downstream compilers analyse and elaborate the selected architecture but cannot silently redesign it, approve procurement, certify installation or manufacture customer commitments.

Example outputs

A single project can produce:

- Reference architecture and topology
- GPU and rack-scale calculations
- NVLink/NVSwitch scale-up architecture
- Spectrum-X / InfiniBand scale-out architecture
- Storage design and sizing
- Optical connectivity requirements
- Preliminary BOM and component quantities
- Vendor implementation options and alternatives
- Security architecture and zones
- Architecture risk register
- Requirements traceability matrix
- Project delivery plan
- Architecture diagrams
- Executive architecture brief
- Full Solution Architecture Definition

Current development direction

Near-term expansion includes **additional NVIDIA, AMD, Cisco, Huawei, Intel platforms.**

Product principle

Neon AI Factory Architect is intended to accelerate delivery of solution architecture by making AI infrastructure architecture reproducible, explainable and commercially usable, while preserving the distinction between architectural engineering, vendor choice, procurement and implementation authority.

Neon AI Factory Architect does not rely on Generative AI, however the option to enable agentic A.I as a decision support integration is provided in current and future versions of the suite.

Target use cases: NeoClouds • GPUaaS • Sovereign AI • Enterprise AI Infrastructure • HPC/AI clusters • RFI/RFP solution development • Architecture assurance